Hints: Probability – 1. Expectation, indicators and linearity

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Linearity: $\mathrm{E}(X+Y) = \sum_\omega (X(\omega)+Y(\omega))\Pr(\omega)$ on a discrete space (in general, by linearity of the integral); no independence is used. Tail formula: $X = \sum_{k\geqslant 1} \mathbf{1}_{\{X\geqslant k\}}$ for integer $X \geqslant 0$, and $X = \int_0^\infty \mathbf{1}_{\{X>t\}}\,dt$ in general (take expectations and swap sum/integral with expectation, all terms being non-negative). Total expectation: split $X = \sum_i X\mathbf{1}_{B_i}$.

Hint 2. Fixed points: $\sum_i \mathbf{1}_{\{\sigma(i)=i\}}$, each with probability $1/n$, so the expectation is 1. Records: position i is a record iff its value is the largest of the first i, probability $1/i$, so the expectation is $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$.

Hint 3. Cycles: $\#\text{cycles} = \sum_i \frac{1}{L_i}$, where L_i is the length of the cycle containing i. L_i is uniform on $\{1,\dots,n\}$ (build the cycle of i step by step: it closes at step ℓ with probability $\frac{n-1}{n}\cdots\frac{n-\ell+1}{n-\ell+2}\cdot\frac{1}{n-\ell+1} = \frac{1}{n}$), so the expectation is $n\cdot\frac{1}{n}\sum_\ell \frac{1}{\ell} = H_n$.

Problem 2

Hint 1. If X_k is the indicator of heads for C_k, then $(-1)^{\#\text{heads}} = \prod_k (1-2X_k)$. Use independence.

Hint 2. $\mathrm{E}\prod(1-2X_k) = \prod_{k=1}^n \frac{2k-1}{2k+1} = \frac{1}{2n+1}$ (telescoping), and $\Pr(\text{odd}) = \frac{1-\mathrm{E}(-1)^{\#}}{2} = \frac{n}{2n+1}$.

Problem 3

Hint 1. Compute the distribution of the number of hits after n shots for small n ($n = 2, 3, 4$). What do you notice?

Hint 2. Claim: after $n \geqslant 2$ shots the number of hits is uniform on $\{1,\dots,n-1\}$. Induction: $\Pr(k \text{ hits after } n+1) = \frac{1}{n-1}\left(\frac{k-1}{n} + \frac{n-k}{n}\right) = \frac{1}{n}$.

Hint 3. Answer: $\frac{1}{99}$.

Problem 4

Hint 1. Condition on $m = \max X$, which ranges over $k,\dots,k+a$: $\Pr(\max X = m) = \binom{m-1}{k-1}/\binom{k+a}{k}$ and $\Pr(\min Y > m) = \binom{k+n+a-m}{n}/\binom{k+n+a}{n}$.

Hint 2. $\sum_m \binom{m-1}{k-1}\binom{N-m}{n} = \binom{N}{k+n}$ with $N = k+n+a$: count the $(k+n)$-subsets of $\{1,\dots,N\}$ by the position m of their k-th smallest element.

Hint 3. Simplify $\binom{N}{k+n}/\left(\binom{k+a}{k}\binom{N}{n}\right)$; the a-dependence cancels and the probability is $1/\binom{n+k}{k}$ (as for $a = 0$, where $X = \{1,\dots,k\}$).

Problem 5

(a)

Hint 1. Write a recurrence by total expectation over the states "nothing useful yet" and "just saw H". For HH: $E_0 = 1 + \frac{1}{2}E_H + \frac{1}{2}E_0$, $E_H = 1 + \frac{1}{2}E_0$, giving 6. For HT, once an H has appeared you only wait for the first T: $2 + 2 = 4$.

(b)

Hint 1. Count the sequences of length n that stop exactly at time n. For HT: the sequence is $T^a H^b T$ with $b \geqslant 1$, so there are $n-1$ of them.

Hint 2. For HH: the sequence is a string without HH ending in T (or empty), followed by HH; the number of such strings of length $n-2$ is the Fibonacci number F_{n-1} ($F_1 = F_2 = 1$).

Hint 3. So you need $F_{n-1} = n-1$; the only solutions of $F_m = m$ with $m \geqslant 1$ are $m = 1$ and $m = 5$. Answer: $n = 2$ or $n = 6$.